D.Y. PATIL
DEEMED TO BE
UNIVERSITY
RAMRAO ADIK
INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

1. A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is given in the following table.

t (sec)	0	10	20	30	40	50	60	70	80
a (m/s^2)	30	31.63	33.34	35.47	37.75	40.33	43.25	46.69	50.67

Find the velocity of rocket at $t = 80$ sec by
(a) Trapezoidal Rule
(b) Simpsons $\left(\frac{1}{3}\right)^{rd}$ Rule

ns $h = 10$

$y_0 = 30$, $y_1 = 31.63$, $y_2 = 33.34$, $y_3 = 35.47$, $y_4 = 37.75$, $y_5 = 40.33$, $y_6 = 43.25$, $y_7 = 46.69$, $y_8 = 50.67$

(a) Trapezoidal rule

$I = \frac{h}{2}[X + 2R]$ where, X = extreme terms (y_0, y_8)

R = Remaining terms $(y_1, y_2, y_3, y_4, y_5, y_6, y_7)$

we have,

$I = \frac{10}{2}[(30 + 50.67) + 2(31.63 + 33.34 + 35.47 + 37.75 + 40.33 + 43.25 + 46.69)]$

$= 5[80.67 + 536.92]$

$= 3087.95$ m/s

(b) By simpsons 1/3rd rule

$I = \frac{h}{3}[X + 4O + 2E]$ where $X = [y_0, y_8]$, $O = [y_1, y_3, y_5, y_7]$, $E = [y_2, y_4, y_6]$

$$I = \frac{10}{3}\left[(30+50.67)+4(31.63+35.47+40.33+46.69)+2(33.34+37.7+43.25)\right]$$

$$= \frac{10}{3}\left[80.67+616.48+228.68\right] = 3086.1 \text{ m/s}$$

2. Evaluate $\int_0^2 e^x dx$ by trapezoidal rule with 5 subintervals.

Ans Let $\int_0^2 e^x dx$, we have $x_0 = 0$ and $x_5 = 2$

$y = e^x$ and $h = \frac{2}{5} = 0.4$

x	0	0.4	0.8	1.2	1.6	2.0
y	1.0	1.4918	2.2255	3.3201	4.9530	7.3891

$I = \frac{h}{2}[X+2R]$ where $X = (y_0, y_5)$, $R = (y_1, y_2, y_3, y_4)$

$$I = \frac{0.4}{2}\left[(1+7.3891)+2(1.4918+2.2255+3.3201+4.9530)\right]$$

$$= 0.2\,(8.3891+23.9808) = 6.47398$$

3. Evaluate $\int_{0.2}^{1.4} (\sin x - \log e^x + e^x)\,dx$ using
   a) Trapezoidal rule
   b) Simpson's $1/3^{rd}$ rule
   c) Simpson's $3/8^{th}$ rule

Ans $\int_{0.2}^{1.4} (\sin x - \log e^x + e^x)\,dx$

x	0.2	0.4	0.6	0.8	1.0	1.2	1.4
y	3.02	2.79	2.89	3.16	3.55	4.06	4.4

a) Trapezoidal rule,

$I = \frac{h}{2}[X+2R]$ where $X = (y_0, y_6)$, $R = (y_1, y_2, y_3, y_4, y$

$$I = \frac{0.2}{2}[(3.02+4.4)+2(2.79+2.89+3.16+3.55+4.06+4.4)]$$

$$= 4.032$$

(b) By simpsons $1/3^{rd}$ rule we get,

$$I = \frac{h}{3}[X + 2E + 4O] \text{ where } X = (y_0, y_6),\ E = (y_2, y_4),$$

$$O = (y_1, y_3, y_5)$$

$$= \frac{0.2}{3}[(3.02+4.4)+2(2.84+3.55)+4(2.79+3.16+4.06$$

$$= 4.022$$

(c) By simpsons $3/8^{th}$ rule,

$$I = \frac{3h}{8}[X+2T+3R] \text{ where } X = (y_0, y_6),\ T = (y_3),$$

$$R = (y_1, y_2, y_4, y_5)$$

$$I = \frac{3}{8}(0.2)[(3.02+4.4)+2(3.16)+3(2.79+2.89+3.55+4.06)]$$

$$= 4.02075$$

4. Find the value of $\int_0^1 \frac{x^2}{1+x^3}\,dx$ using

(a) Trapezoidal rule,

x	0	0.25	0.5	0.75	1.0
y	0	0.0588	0.2	0.36	0.8

$$I = \frac{h}{2}[X + 2R] \text{ where } X = (y_0, y_4),\ R = (y_1, y_2, y_3)$$

$I = \frac{h}{2}[(y_0+y_4)+2(y_1+y_2+y_3)]$

$= \frac{0.25}{2}[(0+0.5)+2(0.0588+0.2+0.36)]$

$= 0.125(0.5+1.2376) = 0.2171$

b) By Simpsons $1/3^{rd}$ rule

$I = \frac{h}{3}[X+2E+4O]$ where, $X=(y_0, y_4)$, $O=(y_1, y_3)$, $E=(y_2)$

$= \frac{0.25}{3}[(0+0.5)+4(0.0588+0.36)+2(0.2)]$

$= 0.2146$

consider, $\int_0^1 \frac{x^2}{1+x^2}dx = \int_0^1 \frac{x^2+1-1}{1+x^2}dx = \int_0^1 x - \frac{1}{1+x^2}dx$

$= \int dx - \int \frac{1}{\sqrt{1+x^2}}dx = [x]_0^1 - [\tan^{-1}x]_0$

$= 1 - \frac{\pi}{4} = 0.215$

For simpsons $1/3^{rd}$ rule these is 0.215,

5. Evaluate $\int_0^1 \frac{1}{1+x^2}dx$ using

a) Trapezoidal rule
b) Simpsons $1/3^{rd}$ rule
c) Simpsons $3/8^{th}$ rule

Ans $\int_0^1 \frac{1}{1+x^2}dx$,

x	0	1/6	2/6	3/6	4/6	5/6	1
y	1	0.9729	0.9	0.8	0.6923	0.5401	0.5

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(a) By trapezoidal rule

$I = \frac{h}{2}[X + 2R]$ where $X = (y_0, y_6)$, $R(y_1, y_2, y_3, y_4, y_5)$

$I = \frac{h}{2}((y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5))$

$= \frac{1}{2}((1 + 0.5) + 2(0.9729 + 0.9 + 0.8 + 0.6923))$

$= 0.78424$

(b) By simpsons $1/3^{rd}$ rule

$I = \frac{h}{3}[X + 2E + 4O]$ where $X = (y_0, y_6)$

$E = (y_2, y_4)$

$O = (y_1, y_3, y_5)$

$I = \frac{1}{18}[(1.5) + 2(0.9 + 0.6923) + 4(0.9729 + 0.8 + 0.5921)]$

$= \frac{1}{18}$

$= 0.7858$

(c) By simpsons $3/8^{th}$ rule

$I = \frac{3h}{8}[X + 2T + 3R]$ where $X = (y_0, y_6)$

$T = (y_3)$

$R = (y_1, y_2, y_4, y_5)$

$= \frac{3}{8}[(1.5) + 2(0.8) + 4(0.9729 + 0.9 + 0.6923 + 0.5901)]$

$= 0.785395$

6. Using the following data, evaluate $\int_1^7 y\,dx$ by trapezoidal rule.

x	1	2	3	4	5	6	7
y	2.105	2.808	3.614	4.604	5.857	7.451	9.467

Ans. $I = \frac{1}{2}[(2.105+9.467)+2(2.808+3.614+4.604+5.857+7.451)]$

$= 0.5(60.24) = 30.12$

7. Evaluate $\int_0^{\pi} \sin x\,dx$ by trapezoidal rule is 10 subintervals.

Ans.

x	0	$\pi/10$	$2\pi/10$	$3\pi/10$	$4\pi/10$	$5\pi/10$	$6\pi/10$	$7\pi/10$
y	0	0.3096	0.5878	0.8090	0.9511	1	0.9511	0.8090

$I = \frac{h}{2}[X+2R]$ where $X = (y_0, y_{10})$

$R = (y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9)$

we have,

$I = \frac{h}{2}[(y_0+y_{10})+2(y_1+y_2+y_3+y_4+y_5+y_6+y_7+y_8+y_9)]$

$= \frac{0.31416}{2}[(0+0)+2(0.3090+0.5878+0.8090+0.9511+1+0.9511+0.8090+0.5878+0.3090)]$

$= 0.15708 \times 12.6276$

$= 1.9835$

8. Evaluate $\int_1^7 y\,dx$ by a) Trapezoidal rule b) Simpsons $\frac{1}{3}$rd rule c) Simpsons $\frac{3}{8}$th rule

x	1	2	3	4	5	6	7
y	2.157	3.519	4.198	4.539	4.708	4.792	4.835

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Ans a) By trapezoidal method rule,

$I = \frac{h}{2}[X + 2R]$ where, $X = (y_0, y_6)$

$R = (y_1, y_2, y_3, y_4, y_5)$

$= \frac{1}{2}[(2.157 + 4.835) + 2(2.519 + 4.198 + 4.539 + 4.708 + 4.792)$

~~$= 0.5(2.157 + 4.835)$~~

$= 0.5(6.992 + 43.512) = 25.252$

b) Simpsons 1/3 rule

$I = \frac{h}{3}[X + 4O + 2E]$ where $X = (y_0, y_6)$

$O = (y_1, y_3, y_5)$

$E = (y_2, y_4)$

$= \frac{1}{3}[(2.157 + 4.835) + 2(4.198 + 4.708) + 4(3.519 + 4.539 + 4.792)$

$= \frac{1}{3}[6.992 + 51.4 + 17.812] = 25.4003$